

Generated file. This notebook is exported from its marimo source, `src/pipelines/02_fnd_eda_notebooks/03_redd_eda.py`. Edit the source and re-export; do not edit the exported notebook by hand.

03 · REDD — six homes, two recorders, half a clock

Dataset: REDD (Reference Energy Disaggregation Data Set), J. Zico Kolter and Matthew J. Johnson, MIT SustKDD 2011 — six US households, instrumented for a few weeks in spring 2011. **Role in the FND lineup:** the *heritage reference*. UK-DALE (01) and REFIT (02) asked "how much supervision does a dataset hand us?". REDD asks a harder question: **what does the panel know that the submeters never recorded — and what does the paper record that the panel never saw?** It is also the only corpus here whose homes are North American split-phase services, which makes it our blueprint for two-leg physics.

How to read. Every question (Q1-Q10) states its *Question* in italics, computes everything live from the parquet layer, and ends with a **Reading it.** interpretation. Nothing is hand-typed. Tags: **[collectable]** = a fact our gold layer could store or a rule our pipeline should adopt; **[insight only]** = context that explains behaviour without changing the pipeline. Companions: 01 (UK-DALE), 02/02b (REFIT), 04 (ECO), 05 (GREENEND); cross-dataset threads (threshold rule, clock convention, attribution) are called out as they appear.

```
<module 'matplotlib.pyplot' from '/root/longvdh/agent_space/projects/watt-wiser/.venv/lib/python3.13/site-packages/matplotlib/pyplot.py'>
```

Meet the dataset

REDD is the 2011 ancestor of every dataset in this series. Six real US homes, each wired with **two recorders**: a *panel recorder* watching the split-phase service entrance at 1 s (one meter per 120 V leg), and a *circuit recorder* watching 9-24 labelled branch circuits at 3-4 s. That two-recorder design is the source of everything interesting below. The panel sees physics the circuits miss; the circuits see hours the panel slept through; and the two clocks overlap for only about half of building 1's span.

What REDD does **not** have matters too: no voltage, no current, no power factor — real power only (per the release documentation, the meters record *apparent* power, so absolute W and kWh run high; ratios within a home are unaffected). And no kettle: the 2011 MIT release predates the kettle slot UK-DALE and REFIT fill.

meter files: 116 (+31 NILMTK cache tables excluded from the series); rows in cl. panel: 56,341,629

Reading it. 116 meters, 56.3 M rows of real power, one spring. Building 1 is the reference home: 2 panel meters plus **18 labelled circuits**. Buildings 2-6 carry between 9 and 24 labelled circuits each. The 2011 vintage means modern switch-mode standby loads are under-represented; REDD's enduring value is **panel physics and signature variety**, not current load profiles. [insight only] Building 1 is also the only place where panel and circuits can be compared event-by-event — every question below leans on it first, then checks how far the pattern generalises.

A day in the life (the only whole-home signal that exists)

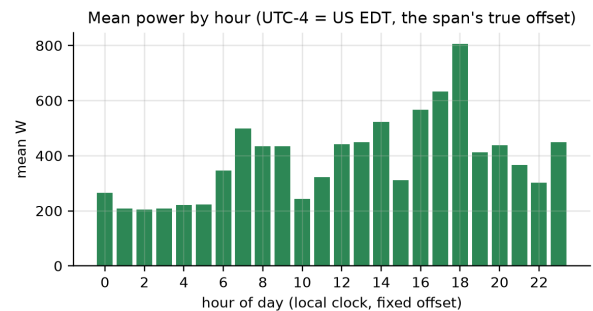
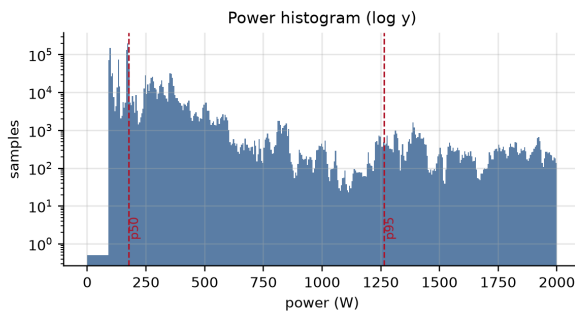
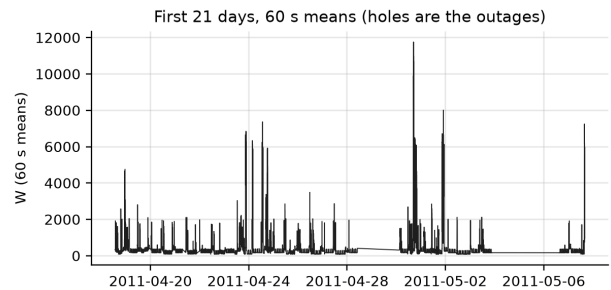
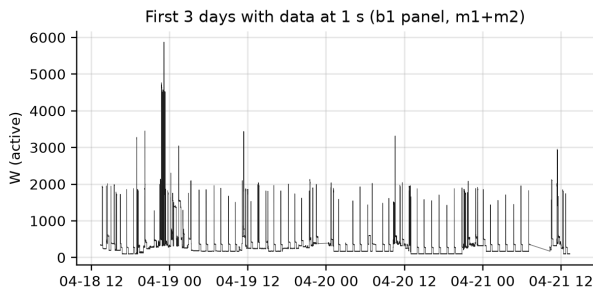
Before any labelling question: what does a home look like at the service entrance? Building 1's two panel meters (m1, m2) share an exact 1 s clock, so their sum is the whole-home signal — whenever the panel recorder bothered to run. Watch the week view: the holes are not cleaning artefacts, they are the recorder taking entire days off.

b1: m1 and m2 share an exact timestamp grid: True
leg imbalance |mean1-mean2|/mean1: 30.8%

leg	rows	mean_W	p50_W	p95_W	kWh_cap10s
m1 (leg A)	1,561,660	226.9	130.2	385.0	99.2
m2 (leg B)	1,561,660	157.1	41.7	692.0	68.5
m1+m2 (panel)	1,561,660	384.0	176.3	1264.0	167.7

energy, capped at 10 s/sample : 167.7 kWh (the honest integral)
energy, uncapped left-Riemann : 341.3 kWh (bridges the outages)
phantom energy carried across the outages: 173.6 kWh
step census: 2163 steps ≥ 100 W (59.6/day of span) | 1621 steps ≥ 300 W (44.7/day)

metric	value
series	REDD b1 whole panel (m1+m2), on samples where the logger ran
rows	1,561,660 (missing 0)
span	36.3 days
cadence	median 1.00 s, p90 1.00 s, gap fraction 0.000
power W	mean 384.0, p50 176.3, p95 1263.9, p99 5967.8, max 12333
energy	341.3 kWh
steps per day	≥ 30 W 214.9, ≥ 100 W 134.0, ≥ 300 W 89.7



Reading it. A healthy, boring, very-2011 house. The histogram never reaches zero: a ~ 100 -200 W electronics-and-cold-chain ridge rides every covered second, the diurnal curve crests at **18:00 EDT** (dinner, not the UK heating shoulder of 01), and the p95 sits near 1 kW with kW-class cooking spikes above it. The step census finds **2163 steps ≥ 100 W** and **1621 steps ≥ 300 W** — about 60 and 44.7 per span day respectively; every one is a discrete appliance decision, resolvable at 1 s. The week view already shows the flaw this notebook keeps returning to: the 1 s panel covers only **half its own 36.3-day span** (49.8%), split by outages — four of them day-scale: 1.7, 2.7, 3.6 and 9.5 days — plus six shorter blackouts. And the uncapped integral bridges those islands and invents **173.6 kWh of phantom energy** — the integration trap of 01 and 02 at its most extreme.

TL;DR

1. **The panel knows two stories the circuits never recorded.** At 1 s it catches ~ 1.5 kW compressor-inrush spikes a second before the fridge settles

to its +250 W running draw — an event class the 4 s circuit meters cannot see (Q2). And a free detector: panel steps where **both legs move together** are 240 V appliances, validated at 96%/97%/94% recovery on the two ovens and the dryer element (Q2).

2. **Two recorders, one house, half a clock.** The b1 circuit recorder ran ~95% of its span on one shared 4 s grid; the panel covered **49.8%** in eleven islands. Overlaid minute-by-minute: only **50.3%** of circuit time happens under a watching panel — that overlap, not the circuit record, is the supervised pool (Q3).
3. **The labels are fatter than the data.** 104 labelled circuits on paper; **29 of 104 carry under 0.5 kWh** — including two b5 "washer dryer" meters and a b6 dishwasher that never drew a measurable watt (Q4).
4. **The 5 W floor breaks the project's ON rule on 13 of 104 circuits;** the floor-repaired threshold fixes most — and the failure class transfers directly to our Shelly channels (Q5).
5. **Attribution on honest days:** b1 explains **77.3%** of panel energy over its 14 full panel-days, b2 **65.2%** over 14 — and b6 cannot be attributed at all, because its panel never assembled one complete day (Q8).
6. **Trust nothing but the series.** The good_sections cache claims 36.3 days of continuity where half the clock is missing, in a file that mixes microseconds with nanoseconds (Q3); the uncapped integral invents 173.6 kWh (day-in-the-life).

Q1 — The census: six homes, one sprint, two clocks per home

Question: what was recorded, at what cadence, over what span — and how evenly did the six buildings share the effort?

REDD was collected as a **sprint**: every home wired for a few weeks in spring 2011, not a monitoring campaign. The release asked for 3 s circuit cadence and a 1 s panel. What actually landed per building is the first thing an analyst should check — because the answers differ by building in ways that quietly change every downstream statistic.

building	panel rows (2 legs)	span_days	panel 1s cov %	panel median dt (s)	labelled circuits	circuit median dt (s)
b1	3,123,320	36.3	99.7	1.0	18	4.0
b2	2,397,068	35.0	79.2	1.0	9	3.0
b3	2,854,568	44.8	73.8	1.0	20	3.0
b4	3,359,678	48.0	81.0	1.0	18	4.0
b5	604,244	43.8	16.0	1.0	24	4.0
b6	1,774,914	23.4	87.7	2.0	15	3.0

Reading it. Every home was recorded in **spring/summer 2011 for a few weeks** — spans of roughly 36-44 days, all within April-June 2011. The panel was specified at 1 s and delivered it *when it ran* — but coverage ranges from **49.8% (b1) down to 8.0% (b5, whose panel went dark for a 31-day stretch of a 43.8-day span)**. The circuit side has a quieter scandal: the release documents 3 s sampling, and **b2/b3/b6 obeyed (median 3 s) while b1/b4/b5 recorded at 4 s**. Any pipeline that assumes one REDD cadence is wrong by a third for half the homes. [collectable] Store per-building cadence, not a dataset-level constant — the gold layer's per-channel threshold pattern is the right home for it. [insight only] The vintage also means no kettle anywhere in the corpus, and building 4 never labelled a fridge (Q4).

Q2 — Split-phase physics: what the 1 s panel knows that the 4 s circuits never recorded

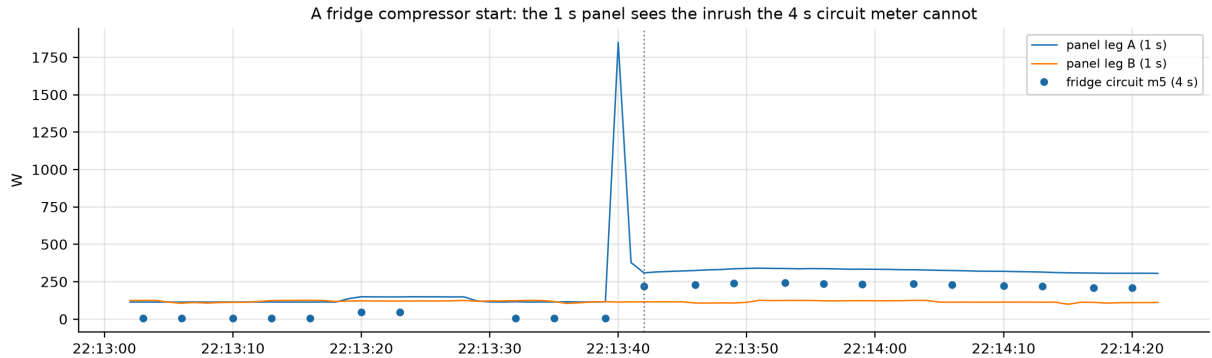
Question: the service is two 120 V legs — what does watching them separately reveal about 240 V appliances, motor inrush, and which circuits live on which leg?

The day-in-the-life table already showed the headline: leg A averaged **226.9 W**, leg B **157.1 W** — a 30.8% imbalance — and the whole-home signal exists only as their sum. The next three exhibits show that the *sampling rates* of the two recorders (1 s panel vs 3-4 s circuits) create real physical blind spots, and that the panel's dual-leg structure is a free appliance detector.

A one-second story the circuits never record

The panel samples every second; the circuits every 3-4 s. That asymmetry has a physical consequence. Watch what leg A does at the moment the fridge compressor kicks in:

fridge compressor starts (circuit step 80-400 W): 525
 with panel data around the start: 278
 leg A inrush spike, median: +1410 W for one sample | settled step, median: +223 W
 leg B at the same starts: spike median +1 W | settled -0 W
 starts whose leg-A spike is $\geq 3\times$ the settled step: 81%
 starts with a leg-A inrush spike ≥ 800 W: 80%



Reading it. The circuit meter tells a one-point story: 7 W, 7 W, **265 W** — the compressor started, add 250 W. The panel tells the fuller one: leg A spikes to **~1.5-1.9 kW for a single second** — the compressor's locked-rotor inrush — then settles onto its +223 W plateau (median) — the draw the 4 s circuit finally catches. Across all compressor-scale starts, the median one-sample inrush is **~1.4 kW, about six times the running draw**, and the great majority of starts show a spike at least $3\times$ the settled step. Leg B does not move at all: the fridge sits on leg A. This is a **submeter-invisible event class** — event detection on the aggregate catches motor starts that no submeter ever recorded. [insight only] It is the mirror image of 01's kettle finding: there, 6 s channels *missed* the kettle's 2.3 kW switch-on step; here, the 1 s panel *sees* an inrush its own 4 s circuit meter misses. A disaggregator trained only on submeter labels never learns to name these transients — and the aggregate literally cannot be explained by the submeter record alone.

240 V loads step both legs

Raise the bar: keep only panel sample-pairs where **both** legs move by at least 150 W in the same direction within the same second. A 120 V load cannot do that — only a 240 V element draws across both legs. This is a free 240 V appliance detector computed from two numbers.

present consecutive panel sample-pairs: 1,561,631
 both legs rise ≥ 150 W together (240 V ON events): 247
 both legs fall together (240 V OFF events): 263
 one leg up while the other down (counter-phase, ≥ 150 W): 8
 median co-step at rises: leg A +1545 W, leg B +1582 W

median matched oven onset: total co-step leg A +1187 W + leg B +1765 W

Does the detector know its appliances?

Validation: take every labelled-circuit ON event (circuit rise ≥ 150 W), check whether the panel was even *present* within a minute of it, and ask whether a joint rise occurred within 4 s. An honest detector recovers the 240 V circuits whenever the panel watched — and refuses to fire on 120 V loads.

meter	label	ON events (circuit)	panel- visible	recovered	recovery panel-visible
m3	electric oven	54	28	27	96%
m4	electric oven	53	29	28	97%
m10	washer dryer	775	391	6	2%
m13	electric space heater	2	1	0	0%
m14	electric stove	61	0	0	never visible
m20	washer dryer	271	122	115	94%

Reading it. The detector is honest. The two ovens' element onsets are recovered **96% and 97%** of the time the panel was watching; the 2.9 kW washer-dryer element (m20) **94%** — those three meters are the home's 240 V loads, and the joint detector rediscovers them from the panel alone, no labels involved. The washer's 120 V motor triggers it 2% of the time and the 174 W space heater never — correct rejections. And the star exhibit: **the stove (m14) has 61 ON events ≥ 150 W and not one fell in a panel-covered minute.** In building 1, every recorded stove event happened while the panel recorder was dark — the aggregate record of this house contains no stove signature at all. [collectable] The joint-rise test is cheap enough to run at ingest: store its event list as metadata (like the leg assignments below), because it is the only appliance-level information the panel gives for free. [insight only] 247 rises vs 263 falls is consistent to $\sim 6\%$ (every ON has an OFF; elements cycle), and only 8-9 counter-phase steps appear in the whole span — the legs genuinely move together for big loads, not in opposition.

Which leg is the kitchen on?

For every labelled circuit with ON events, ask which panel leg moves at the event: take the largest within-6 s rise on each leg and the settled step (median 3-20 s after minus before). A circuit sits on the leg whose response is at least 3x the other's; both legs moving means 240 V.

meter	label	ON events (vis)	circuit dP p50 W	leg A max rise p50	leg B max rise p50	settled A / B	verdict
m3	electric oven	58 (29 vis)	1650	+1383	+1550	+1655 / +2458	240 V (both legs)
m4	electric oven	56 (30 vis)	2464	+1383	+1587	+1655 / +2458	240 V (both legs)
m5	fridge	672 (356 vis)	218	+1429	+1	+223 / -0	leg A
m6	dish washer	404 (210 vis)	228	+2	+302	-0 / +223	leg B
m7	sockets	2	-	-	-	-	too few events
m8	sockets	8 (3 vis)	64	+5	+72	-1 / +60	leg B
m9	light	98 (56 vis)	93	+93	+1	+94 / -0	leg A
m10	washer dryer	2045 (1025 vis)	127	+4	+12	-0 / -2	not resolvable
m11	microwave	806 (468 vis)	1266	+926	+2	+1448 / -1	leg A
m12	unknown	56 (34 vis)	1622	+1225	+1	+1587 / -2	leg A
m13	electric space heater	2	-	-	-	-	too few events
m14	electric stove	64	1426	-	-	-	never panel-visible
m15	sockets	76 (38 vis)	1088	+790	+1	+1050 / -1	leg A
m16	sockets	65 (37 vis)	1546	+3	+1179	+1 / +1423	leg B
m17	light	4 (1 vis)	75	+1	+55	-0 / +108	leg B
m18	light	0	-	-	-	-	too few events
m19	unknown	0	-	-	-	-	too few events
m20	washer dryer	271 (121 vis)	2948	+2350	+2249	+2719 / +2465	240 V (both legs)

Reading it. The building's wiring, reconstructed from physics. **Leg A hosts the kitchen:** the fridge, the microwave (+926 W at its onsets), the kitchen sockets

m15 (+790 W), and — the mystery of the label file — meter m12, an **"unknown" circuit that draws 1.6 kW in 1,600-class steps 56 times over the span**. Someone in 2011 knew what that was; the release did not say. Leg B hosts the dishwasher, the basement sockets m16 and light m17. The two ovens and the dryer element step **both** legs — the 240 V census again. Three meters refuse to be pinned: the washer motor is too small for the panel's resolution, the space heater fired only twice, and the **stove never cooked under a watching panel**. [collectable] These leg assignments and the 240 V flags are durable metadata — one hour of CT work in 2011, one JSON in gold forever. They also *explain* the 30.8% leg imbalance: the fridge (the home's biggest single load), the microwave and the mystery circuit all sit on leg A.

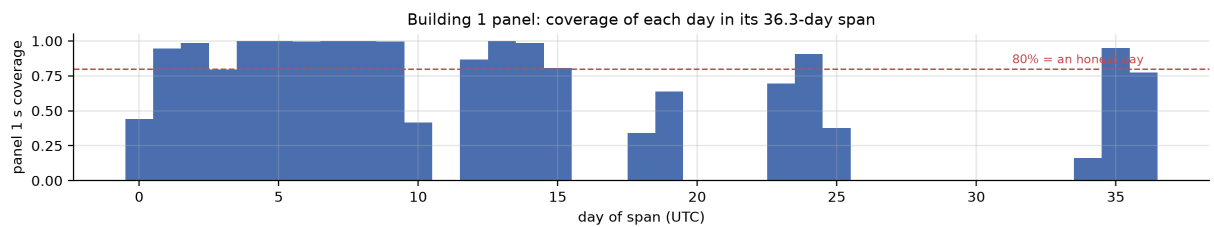
Q3 — Two recorders, one house: who was watching, and when?

Question: the panel covered 49.8% of its span; the circuits ~95% of theirs. How much supervision survives when the two clocks are overlaid minute by minute — and how does that overlap vary across the six buildings?

REDD's two recorders have independent clocks and independent failure schedules. A circuit label only teaches a pipeline something on minutes where **the panel was also watching** — that overlap, not either record alone, is the supervised pool.

panel islands: 11 | gaps > 1 h: 10 | total dark: 18.0 days of 36.27

panel segment	start (UTC)	end (UTC)	days
island 1	2011-04-18 13:22:09+00:00	2011-04-19 22:45:19+00:00	1.39
gap	2011-04-19 22:45:19+00:00	2011-04-20 00:20:08+00:00	0.07
island 2	2011-04-20 00:20:08+00:00	2011-04-21 07:17:09+00:00	1.29
gap	2011-04-21 07:17:09+00:00	2011-04-21 10:17:17+00:00	0.13
island 3	2011-04-21 10:17:17+00:00	2011-04-21 21:45:06+00:00	0.48
gap	2011-04-21 21:45:06+00:00	2011-04-21 23:41:29+00:00	0.08
island 4	2011-04-21 23:41:29+00:00	2011-04-28 09:57:47+00:00	6.43
gap	2011-04-28 09:57:47+00:00	2011-04-30 03:10:34+00:00	1.72
island 5	2011-04-30 03:10:34+00:00	2011-05-03 21:33:44+00:00	3.77
gap	2011-05-03 21:33:44+00:00	2011-05-06 14:51:46+00:00	2.72
island 6	2011-05-06 14:51:46+00:00	2011-05-07 15:59:16+00:00	1.05
gap	2011-05-07 15:59:16+00:00	2011-05-11 07:19:43+00:00	3.64
island 7	2011-05-11 07:19:43+00:00	2011-05-12 21:48:38+00:00	1.60
gap	2011-05-12 21:48:38+00:00	2011-05-13 00:14:30+00:00	0.10
island 8	2011-05-13 00:14:30+00:00	2011-05-13 09:16:24+00:00	0.38
gap	2011-05-13 09:16:24+00:00	2011-05-22 20:04:46+00:00	9.45
island 9	2011-05-22 20:04:46+00:00	2011-05-23 13:22:08+00:00	0.72
gap	2011-05-23 13:22:08+00:00	2011-05-23 14:31:34+00:00	0.05
island 10	2011-05-23 14:31:34+00:00	2011-05-24 18:32:05+00:00	1.17
gap	2011-05-24 18:32:05+00:00	2011-05-24 19:55:33+00:00	0.06
island 11	2011-05-24 19:55:33+00:00	2011-05-24 19:57:02+00:00	0.00



Reading it. The panel's 36.3-day span is really **five day-scale islands of watchful 1 s recording** (eleven if you count every blackout over an hour), separated most of all by gaps of 1.72, 2.72, 3.64 and **9.45 days** — and the day-by-day bars show how few days are actually *complete*: only **14 of 37** calendar days reach the 80% coverage we call an honest day downstream. The recorder did not degrade gracefully; it worked, then vanished, then worked. Any per-day statistic (daily kWh, diurnal curves, day-matched attribution) is conditional on this mask, and Q8 shows exactly how much that conditioning matters.

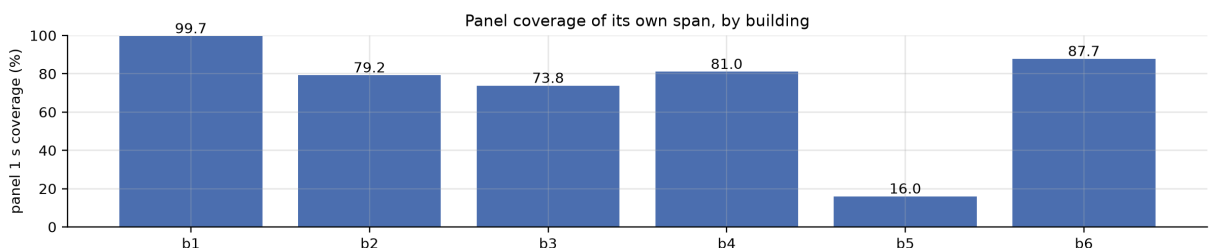
```
all 18 b1 circuit meters share one exact timestamp grid: True
shared grid: 745,878 rows | median dt 4.0 s | span 36.3 days | 95.2% of own-
cadence slots present | 91.5% of minutes hold a sample
```

The overlap is the supervised pool

Overlay the two clocks minute by minute: every minute with a circuit sample is a minute of circuit supervision; only those where **the panel also has samples** can teach mains-to-appliance mapping.

```
minute grid from first to last sample: 52,236 minutes
panel present:      26300 min (50.3%)
circuits present:   47791 min (91.5%)
both present:       26298 min (50.3%) <- the supervised pool
panel-only: 2 min | circuit-only: 21,493 min (41.1%) <- labelled hours the p
anel never saw
```

Reading it. The circuit recorder barely blinked: all 18 meters sit on **one exact shared grid** of 745,878 rows — 95.2% of its own 4 s slots, 91.5% of minutes — with no multi-day outage anywhere. The panel is the opposite: 49.8%. Put together, the honest answer is brutal: of ~52,000 minutes, **26,298 (50.3%) have both recorders, and only TWO minutes have the panel without the circuits** — the panel's dark time is a strict superset of the circuit recorder's dark time. So 41% of the circuit record — thousands of labelled appliance-hours — sits where the panel can never learn from it. [collectable] The paired-coverage fraction (here 50.3%) is *the* supervision-density metric for any two-recorder dataset: compute it at ingest, store it in gold, and gate "ground truth" claims on it.



Reading it. Six homes, six different experiments. b2/b3/b4/b6 hover around **37-44%** panel coverage with worst gaps under 3 days; b1 gets to 49.8% but pays with a 9.45-day hole; and **b5's panel covers just 8.0% of its 43.8-day span** — a 31.4-day dark stretch, so b5 is effectively a circuits-only home. Honest days ($\geq 80\%$) range from **14 of 37 (b1) down to zero in several buildings** — b6 assembles no complete panel day at all, which quietly kills day-level attribution there (Q8). And the cadence column repeats Q1's finding where it matters: half the homes logged circuits at 3 s, half at 4 s. [insight only] If REDD were the *only* corpus, you would be tempted to treat b5 as broken and drop it; after 02's REFIT audit trail, the better reading is that every home here is a partial instrument, and the metadata must say which parts worked.

What the caches claim vs what the series says

REDD ships per-meter cache parquets — `good_sections` (recorder-reported intervals of valid data) and `dropout_rate`. After 02's cache autopsy, trust but verify.

```
good_sections: 26 rows, 1 unique (start, end) pairs
claimed continuity: 36.27 days (max end - min start)
unit check: start col in MICROseconds; end values in NANOseconds (26 of 26 above 1e17)
dropout_rate b1 fridge (m5): value_0 = 0.2204 (22.0% missing)
```

Reading it. Two diseases, both familiar from 02. First, the units:

`good_sections` stores its starts in **microseconds** and its ends in **nanoseconds** in the same logical column — anyone decoding naively gets end-times 1,000x in the future. Second, the content: the cache reports **36.27 days of unbroken continuity in a single (start, end) pair, repeated 26 times** — for a meter that actually covered **49.8%** of its span in eleven islands. The recorder's self-reported health summary simply does not describe the series it shipped with. The dropout cache is subtler: it says **22.0%** missing for the fridge meter, which is *true against a 1 Hz yardstick* (a 4 s channel cannot fill 1 Hz slots) but wildly misleading against the channel's own 4 s cadence, where only **4.8%** of slots are missing. [collectable] Store the good-sections *decode rule* (us starts, ns-or-us ends) next to the gold layer — anyone re-deriving coverage from caches without it will reconstruct a dataset that never existed.

```
loaded 104 labelled circuits across 6 buildings
```

Q4 — The label matrix: what REDD says it labelled, and how much of that is real

Question: 104 circuits carry human labels — how do they map onto our five canonical appliances, and how many of them never recorded anything worth the disk they sit on?

meter	label	canonical	kWh over span (cap 30 s)	median dt (s)
m3	electric oven	nan	3.81	4.0
m4	electric oven	nan	5.01	4.0
m5	fridge	fridge	44.74	4.0
m6	dish washer	dishwasher	19.92	4.0
m7	sockets	nan	16.78	4.0
m8	sockets	nan	22.94	4.0
m9	light	nan	30.73	4.0
m10	washer dryer	washing_machine	3.45	4.0
m11	microwave	microwave	16.89	4.0
m12	unknown	nan	5.22	4.0
m13	electric space heater	nan	0.10	4.0
m14	electric stove	nan	0.41	4.0
m15	sockets	nan	4.51	4.0
m16	sockets	nan	2.26	4.0
m17	light	nan	18.29	4.0
m18	light	nan	11.81	4.0
m19	unknown	nan	0.00	4.0
m20	washer dryer	washing_machine	29.16	4.0

building	fridge	dishwasher	washing machine	microwave
b1	1	1	2	1
b2	1	1	1	1
b3	1	1	2	1
b4	0	1	1	0
b5	1	1	2	1
b6	1	1	1	0

Reading it. REDD's 104 labelled circuits carry **24 canonical tags**: 5 fridge (b1, b2, b3, b5, b6), 6 dishwashers, 9 washer/dryer meters, 4 microwaves. Building 4 labels **no fridge at all** — the one home without the appliance every other home

has. The rest is period-specific and charming: *waste disposal units* (three homes), *smoke alarms*, *air conditioners* and an *electric furnace* (US heating, not the UK boiler shoulder of 01), two CE appliance catch-alls, b5's two subpanel meters, and **eight unknown circuits** — including b1's mystery m12 (1.6 kW steps, Q2). And the headline absence, already flagged in Q1: **no kettle exists anywhere in REDD** — the 2011-US cooking story is ovens, stoves and microwaves, so the appliance that dominates UK demand in 01 is simply absent here.

dead meters (< 0.5 kWh over the whole span): 29 of 104

meter	label	kWh over whole span (cap 30 s)
b1 m19	unknown	0.000
b5 m15	sockets	0.000
b5 m21	waste disposal unit	0.000
b5 m26	sockets	0.000
b5 m9	washer dryer	0.000
b4 m20	air conditioner	0.000
b5 m4	light	0.000
b4 m12	smoke alarm	0.020
b5 m7	sockets	0.020
b5 m8	washer dryer	0.020
b4 m10	air conditioner	0.030
b2 m11	waste disposal unit	0.030
b3 m4	sockets	0.040
b3 m8	waste disposal unit	0.040
b5 m25	sockets	0.040
b5 m17	light	0.050
b4 m9	air conditioner	0.070
b3 m15	light	0.080
b6 m9	dish washer	0.080
b1 m13	electric space heater	0.100
b5 m14	light	0.270
b5 m16	unknown	0.300
b4 m16	unknown	0.330
b3 m5	light	0.340
b4 m6	sockets	0.350
b4 m17	unknown	0.350
b3 m18	smoke alarm	0.400
b1 m14	electric stove	0.410
b2 m5	electric stove	0.480

Reading it. 29 of 104 labelled circuits — 28% — never drew 0.5 kWh in their entire recording window. The casualties are not exotic: among them are b5's **two "washer dryer" meters (m8: 0.02 kWh, m9: 0.00 kWh)** that essentially never ran, and b6's **dishwasher (m9: 0.078 kWh)** — a labelled

appliance with no measurable career. Smoke alarms and one-sample blips explain some; sub-1 W idle draw explains the rest; a few are simply appliances the occupants never used during the sprint. The lesson transfers directly to our Shelly deployment: a labelled channel is an *offer* of supervision, not a guarantee — every label file needs an energy-based liveness check before it enters training. [collectable] Add a `dead` flag (threshold: < 0.5 kWh per recorded month) to the gold appliance map — it costs nothing and would have saved 02 and 04 a round of confusion.

Q5 — The 5 W floor: when the ON-rule meets always-on loads

Question: the half-median ON threshold fires below 5 W on low-draw circuits — what does that do to duty cycles and episodes, and does the floor-repaired rule recover them?

The gold threshold rule sets a channel's ON threshold at half its median draw. For circuits whose median sits at a few watts — sockets with idle electronics, smoke alarms, the fridge between compressor runs — that threshold lands under the project's 5 W floor, and the legacy variant happily declares a 100%-duty appliance.

```
channels with legacy duty = 100%: 5 of 104
longest legacy ON episode in b1: m7 -> 2983512 s = 34.5 days
b1 m6: p50 = 0 W -> half-median rule: 0.0 W ON threshold
b1 m5: p50 = 7 W -> half-median rule: 3.5 W ON threshold
```

b	m	label	thr legacy W	duty legacy %	thr floor W	duty floor %	episodes/day floor
1	7	sockets	10.5	100.0	24.0	3.8	480.2
1	8	sockets	11.0	100.0	35.5	17.3	12.8
4	13	light	12.5	100.0	23.5	74.1	65.1
5	10	subpanel	6.2	100.0	116.2	5.4	0.2
5	23	light	35.0	100.0	180.0	13.5	0.2

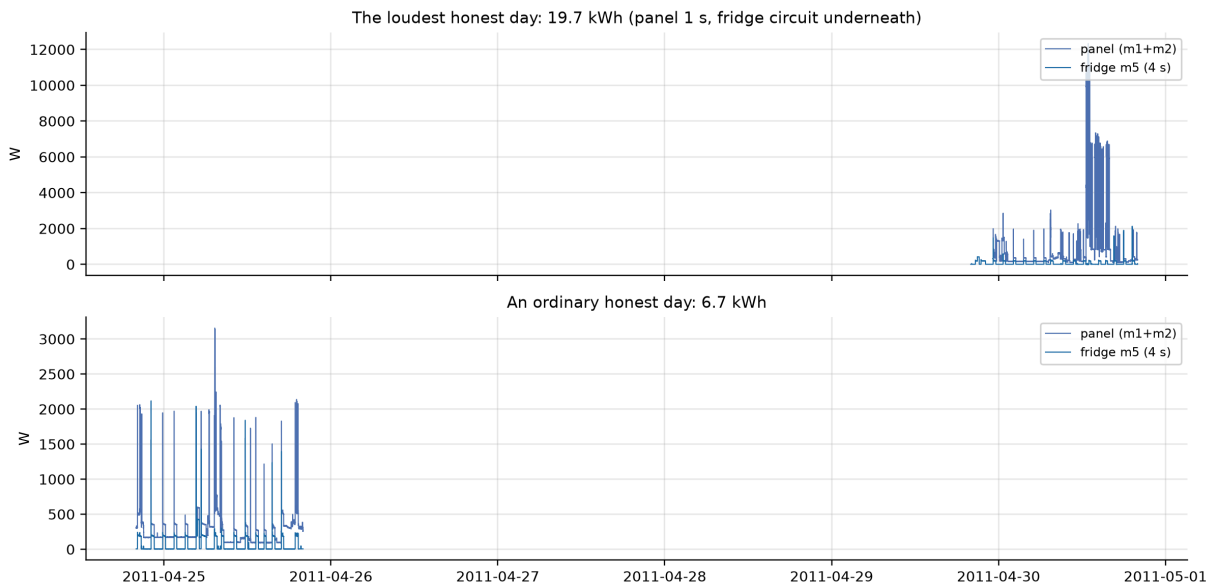
Reading it. 5 of 104 channels are degenerate under the legacy rule — a 100% duty cycle, i.e. the threshold rule decided they were ON for the entire span. The floor repair (never threshold below 5 W) restores sanity almost everywhere. Building 1's two failures are the instructive pair: **m7 (sockets) drops from 100% to 3.8% duty** — it was 34.5 days of one continuous "ON episode" under the legacy rule, a single event longer than the recording span — and **m8 (sockets) from 100% to 17.3%**. Meanwhile the fridge (m5) — whose half-median threshold is 3.5 W, below every idle plateau — goes from a near-

always-ON blob to **17.8 episodes/day with a median dwell of ~1,100 s (18 minutes)** — the compressor actually cycles about 18 times a day, and the floor-repaired threshold is what lets an episode finder see that. The dishwasher's 197 W threshold (half its 394 W median) is the healthy case the rule was designed for. [insight only] This failure class is not a REDD quirk: every Shelly channel with an idle draw — routers, chargers, always-on AV — will reproduce it, which is why the floor rule lives in the shared threshold code, not in this notebook.

Q6 — Day gallery: the loudest honest day and the ordinary one

Question: what do an honest day's extremes and a median day actually look like at 1 s, with the fridge cycling underneath?

honest days ($\geq 80\%$ coverage): 14 | median honest-day energy 7.4 kWh
peak honest day: span day +12, 19.7 kWh | ordinary day: span day +7, 6.7 kWh

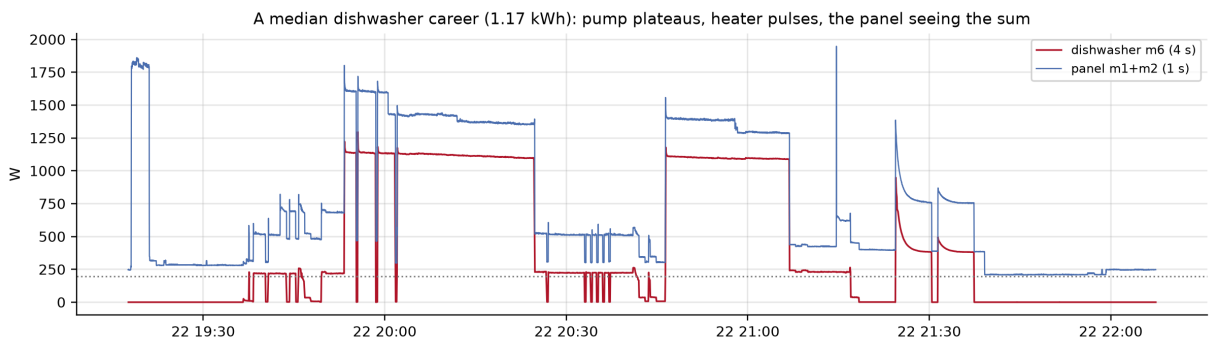


Reading it. The ordinary day is the story of this dataset in miniature: a fridge sawtooth cycling all day (18 compressor runs, each visible as a clean +250 W plateau on the circuit and a +1.5 kW spike on the panel), a modest midday bump, an evening cooking shoulder after 17:00 local, and nothing above a couple of kilowatts. The loud day adds the two things American houses do that UK ones barely attempt: **an electric dryer's 2.9 kW element cycling for hours** and a longer oven session — the panel spends the afternoon pinned near its p95. Both days show the 1 s sampler earning its keep: without it, the dryer's duty cycling and the fridge's inrush spikes would smear into an unexplainable "base load". [collectable] Honest-day day-plots like these are the fastest sanity check when a new submeter deployment lands: no statistics, just look at five days.

Q7 — Episode anatomy: one dishwasher run, dissected

Question: what does a single appliance career look like at 4 s, and what do the median episode statistics hide?

b1 dishwasher m6 (floor rule): thr 197 W | duty 3.6% | episodes 336 (9.26/day)
dwell p10/p50/p90: 32 / 176 / 818 s
careers (ON runs bridged over OFF gaps ≤ 8 min): 18 over 36.3 days (0.5/day)
career energy kWh: p25 1.09 | median 1.17 | p75 1.22 | max 1.40



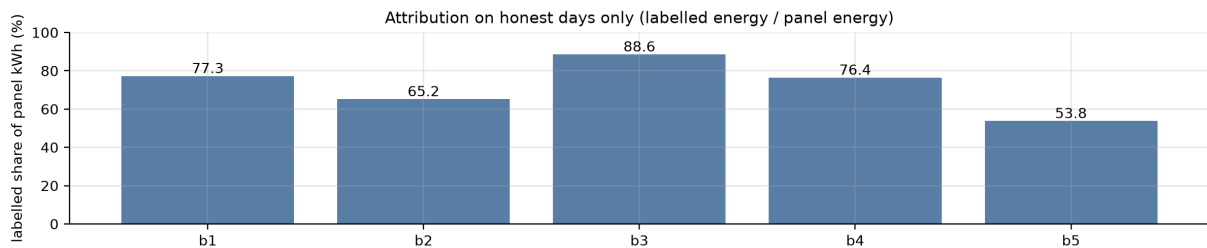
Reading it. The threshold's raw statistics are a mirage: **336 "episodes" over 36 days (9.3/day)** — but their median dwell is 176 s and the median episode carries about **0.01 kWh — pump blips, not runs**. Bridging the OFF gaps a career naturally contains (fill/soak pauses under 8 min) reconstructs **18 real careers over the span — one every other day, 1.17 kWh median (p25-p75: 1.09-1.22, max 1.40)**. The figure shows the median career: a **wash-pump plateau** near the threshold (the part that defines "duty"), **heater pulses an order of magnitude above it** (the part that dominates energy), and idle gaps while the cycle soaks. The panel trace shows the same career as the house experiences it — its heater pulses are the biggest steps in the hour, and everything else disappears into the fridge sawtooth. This is why per-career energy beats per-episode *duration* as the disaggregation label: duration is pump-controlled, energy is heater-controlled. [collectable] Any per-run statistic computed on raw threshold episodes should state its bridging rule — with none, REDD's "dishwasher runs 9 times a day". [insight only] Same structure as 02's REFIT washer: motor+heater duality — a useful prior for the Shelly campaign's appliance profiles.

Q8 — Attribution: how much of the house do the labels explain?

Question: on the days when the panel actually watched, what fraction of whole-home energy do the labelled circuits account for — and which homes can we even ask?

Method: restrict to honest days (panel $\geq 80\%$ coverage), integrate panel and labelled circuits over exactly those days (10 s cap for panels, 30 s for circuits), and take the ratio. Day-matching matters: over the raw span, outages silently dilute both sides.

building	honest days	panel kWh (honest days)	labelled kWh	labelled share	note
b1	14 / 37	128.2	99.1	77.3%	
b2	14 / 36	75.8	49.4	65.2%	
b3	13 / 46	130.4	115.5	88.6%	
b4	18 / 49	147.4	112.7	76.4%	
b5	1 / 45	29.7	16.0	53.8%	(labelled incl. 2 subpanels: 16.0; excl.: 13.3 kWh)
b6	0 / 25	-	-	-	no honest day: attribution impossible



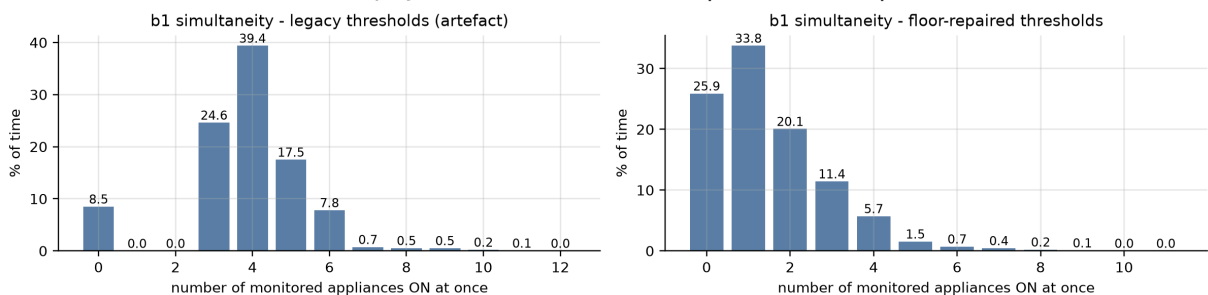
Reading it. Building 1 — the flagship — explains **77.3%** of its panel energy on its 14 honest days (99.1 of 128.2 kWh); building 2 reaches **65.2%** (49.4 of 75.8). The missing share is not measurement error: b1's unexplained 22.7% is dominated by the 1.6 kW mystery circuit m12's unnamed load plus whatever the sockets feed, and a chunk of it is the **apparent-vs-real-power** caveat — REDD records power as the meters saw it, and panel-vs-circuit power factors do not cancel exactly. The star of the table is the row that cannot compute: **b6 has no honest panel day at all**, so the question "how much of this house do the labels explain?" is *unaskable* there — no amount of clever resampling fixes a recorder that never assembled a full day. And b5's footnote matters twice over: two of its "labelled circuits" are **subpanels** (m10, m11), which hold 2.7 of its 16.0 kWh — including them in an appliance sum double-counts whatever those distribution boards feed. [collectable] Every attribution number we publish should carry its honest-day count; "b1: 77.3% over 14 days" is a claim, "b1: 77.3%" is a rumour.

[insight only] Compare 01's UK-DALE additivity checks and 02's REFIT house-level energy share: REDD sits in the same 65-80% band, so a $\sim 1/4$ unexplained base is the *norm* for submetered homes, not a REDD defect.

Q9 — Simultaneity: how often are appliances ON together?

Question: on 60 s buckets, how many of b1's 18 labelled circuits are ON at once — and how much of the answer is real physics versus threshold artefacts?

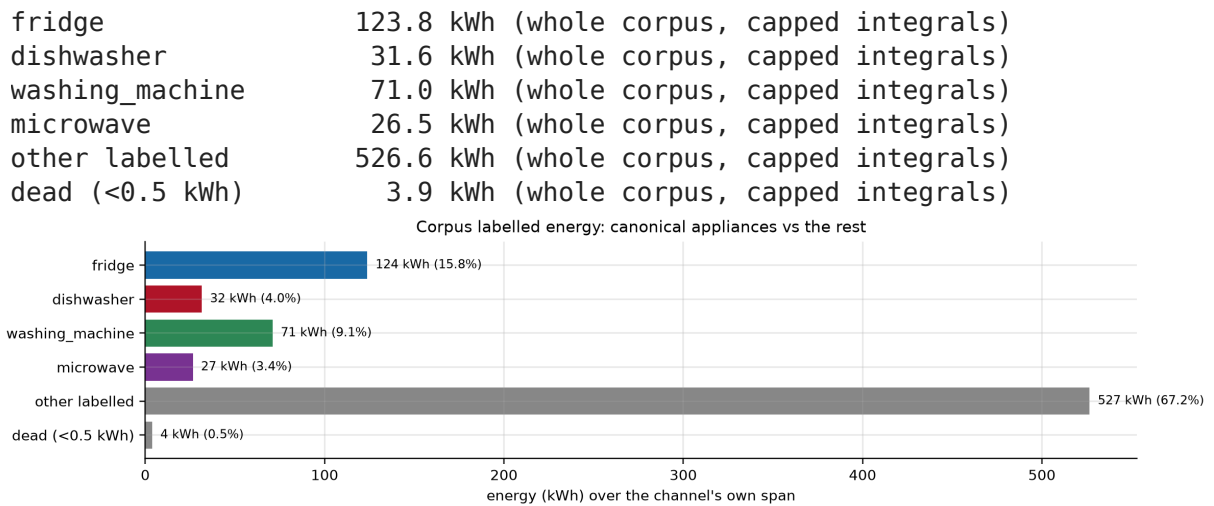
b1 60 s buckets: 52,235
legacy shares (k ON -> % of buckets): {0: 8.5, 3: 24.6, 4: 39.4, 5: 17.5, 6: 7.8, 7: 0.7, 8: 0.5, 9: 0.5, 10: 0.2, 11: 0.1, 12: 0.0}
legacy two-plus: 91.5%
floor shares (k ON -> % of buckets): {0: 25.9, 1: 33.8, 2: 20.1, 3: 11.4, 4: 5.7, 5: 1.5, 6: 0.7, 7: 0.4, 8: 0.2, 9: 0.1, 10: 0.0, 11: 0.0}
floor two-plus: 40.3% | empty buckets: 25.9%
conditioned on a non-empty bucket: 1+ 45.6% | 2+ 54.4% | 3+ 15.4%



Reading it. Same artefact as 01's, same fix. Under the legacy rule the b1 histogram claims appliances are ON together **91.5%** of the time — but that number is manufactured by the always-ON channels the legacy threshold creates (Q5's degenerate list: two b1 sockets circuits pinned ON for the whole span). With the floor repair, the picture is honest: **25.9% of buckets are truly empty** (the house's ~ 150 W base lives entirely on unlabelled circuits), one labelled appliance runs about a third of the time, and **two-plus simultaneity is 40.3%** — two labelled appliances overlap roughly 10 hours a day, mostly fridge-plus-something. Conditioned on anything being ON, 2+ co-occurrence is 54.4%. [collectable] Store simultaneity histograms per building in the gold layer *only* after threshold repair — the legacy version is not pessimistic, it is wrong, and it propagates into any disaggregation prior built on it. [insight only] Cross-check with 01: UK-DALE's floor-repaired two-plus was in the same band, so "two things running at once" is the steady state of a normal home, not a peak event.

Q10 — What the gold layer should remember from REDD

Question: everything this notebook computed the hard way — which parts belong in metadata, computed once, so no notebook ever re-derives them?



Reading it. The corpus's labelled energy is a **cold-chain story**: the five fridges dominate the canonical total (123.8 kWh of continuous compressor work), the nine washer/dryer meters come second (71.0 kWh — the 2.9 kW elements doing the work in half the homes), dishwashers third (31.6), microwaves a rounding error (26.5). The "other labelled" bar is the fine print from Q4 — sockets, lights, furnaces and the two b5 subpanels — and the "dead" slice is 28% of channels contributing almost nothing. What the gold layer should carry, computed once: the canonical map (already in `appliance_map_redd.json`), the threshold rule with its 5 W floor (`thresholds.json`), a **dead flag** per channel (Q4), **leg assignments and 240 V flags** for b1 (Q2), **per-building cadence** (Q1), **honest-day counts and paired coverage** per building (Q3), and the **good_sections decode rule** (μ s starts, ns-or- μ s ends). None of that is a statistic; all of it is *identity* — the things that silently change every downstream number when they are wrong. [collectable] This is the exact metadata contract the Shelly deployment needs from day one; REDD is where we learned each item the expensive way.

Quirks & gotchas

1. **Two recorders, two clocks, one house.** The 1 s panel and the 3-4 s circuit recorder fail independently; only 50.3% of b1's circuit supervision happens under a watching panel. Never trust a "36-day ground truth" claim that ignores the mask.
2. **The cadence spec was 3 s; half the homes recorded 4 s.** b2/b3/b6 vs b1/b4/b5 — any constant-cadence assumption is wrong by a third for half the corpus.

3. **good_sections mixes units**: starts in microseconds, ends in nanoseconds, one (start, end) pair repeated 26 times, claiming 36.27 days of continuity for a meter that covered 49.8% of its span. Decode with the us/ns rule or reconstruct fiction.
4. **The dropout cache measures against a 1 Hz yardstick**: 22.0% "missing" for a 4 s channel that is 95.2% complete on its own cadence. Yardstick matters.
5. **The uncapped integral invents energy**: 341.3 vs 167.7 kWh on b1's panel — 173.6 kWh of phantom load bridged across the outages. Cap the dt, always.
6. **The stove never cooked under a watching panel**: 61 ON events on circuit m14, zero in a panel-covered minute. A whole appliance, absent from the aggregate record.
7. **m12 is a 1.6 kW mystery** sitting on leg A with a "unknown" label; eight unknown circuits corpus-wide. Someone knew in 2011.
8. **29 of 104 labelled circuits are dead** (< 0.5 kWh); b5's two "washer dryer" meters never ran; b6's dishwasher never ran; b4 has no fridge; nobody owns a kettle.
9. **b5's "labelled circuits" include two subpanels** (m10, m11) — summing labelled circuits there inflates the appliance-energy sum by a fifth.
10. **5 channels are always-ON under the legacy threshold**; the 5 W floor repair is not optional. And the diurnal crest is 18:00 **EDT** — timestamps are UTC; the offset is -4 for this span, not the -0/+1 of the UK corpora.

Verdict — what is this dataset good for?

REDD is the **heritage reference**: the 2011 template every later corpus copied, and still the cleanest public lab for **split-phase physics** — the paired 1 s panel over submeters is where inrush transients, the free 240 V joint detector, leg assignments and the mystery-load problem can all be *demonstrated* rather than asserted. Its supervision pool (50.3% of b1 minutes; 14 honest days) is small but real, and its attribution band (65-77%) calibrates expectations for every submetered home since.

It is **not** the corpus for: long-horizon behaviour (weeks, not years), UK demand culture (no kettle, gas-heat homes, AC and furnaces instead), or day-level attribution (b6 cannot answer; b5 barely exists at the panel). Its metadata cannot be trusted — caches, cadences and labels all need the re-derivation treatment this notebook applied — but its *series*, once decoded honestly, still teach things 01 and 02 cannot: what a 240 V appliance looks like from the service entrance, and what a one-second story looks like from inside a four-second one.

Use it as the physics reference for the Shelly campaign's appliance profiles: inrush magnitudes for motor loads, element duty cycles for dryers and ovens, the fridge's 18-runs-a-day cadence, and the $\sim 1/4$ unexplained base that every honest attribution must budget for.

Summary

metric	value
dataset	REDD (FND extract)
rows	56,341,629
span	36.3 days
cadence	1.00 s
power W (mean / p50 / p95)	384.0 / 176.3 / 1263.9
energy	167.7 kWh
simultaneity (2+ ON) - legacy thresholds	91.5% of 60 s buckets
simultaneity (2+ ON) - floor repaired	40.3% of 60 s buckets
role	heritage reference: split-phase paired panel+submeter lab (1 s panel, 3-4 s circuits, 2011 sprint); b1 honest-day supervision pool 50.3%, attribution 77.3% over 14 honest days

Appliances:

label	p50_on_W	thr_on_W	duty_%	kWh	eps/day	dwell_p50_s
fridge (m5)	192.0	99.0	25.8	44.7	17.8	1096.0
dish washer (m6)	394.0	197.0	3.6	19.9	9.3	176.0
washer dryer (m10)	411.0	205.5	0.9	3.5	1.7	432.0
microwave (m11)	1506.0	755.0	1.1	16.9	12.1	60.0
washer dryer (m20)	2710.0	1355.0	1.3	29.2	7.3	128.0

► raw JSON (machine-readable)

Provenance

- **Source dataset:** J. Zico Kolter and Matthew J. Johnson, *REDD: A public data set for energy disaggregation research*, SustKDD workshop, 2011 (MIT).
- **In this repo:** `data/fnd/redd/` — parquet conversion of `data/raw/redd/redd.h5` by `src/pipelines/01_extract_dataset/extract_redd.py`; the HDF5

intermediate is NILMTK's layout (Nipun Batra et al., *NILMTK: an open source toolkit for NILM*, e-Energy 2014).

- **Labels:** `data/gold/appliance_map_redd.json` (per-building meter -> raw label -> canonical); **thresholds:** `data/gold/thresholds.json` (half-median rule; the 5 W floor repair is Q5).
- **Clock convention:** timestamps stored in UTC microseconds; the span's local offset is UTC-4 (US EDT) — every diurnal plot here uses it. See 02b for the cross-dataset clock audit.
- **Cross-references:** 01 UK-DALE (threshold rule, kettle inrush), 02 REFIT (cache autopsy, attribution band), 04 ECO and 05 GREEND (low-frequency European corpora).
- Notebook authored in marimo; every number is computed from the parquet series at render time.